VELS
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PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR - CHENNAI

Department of Artificial Intelligence and Machine Learning

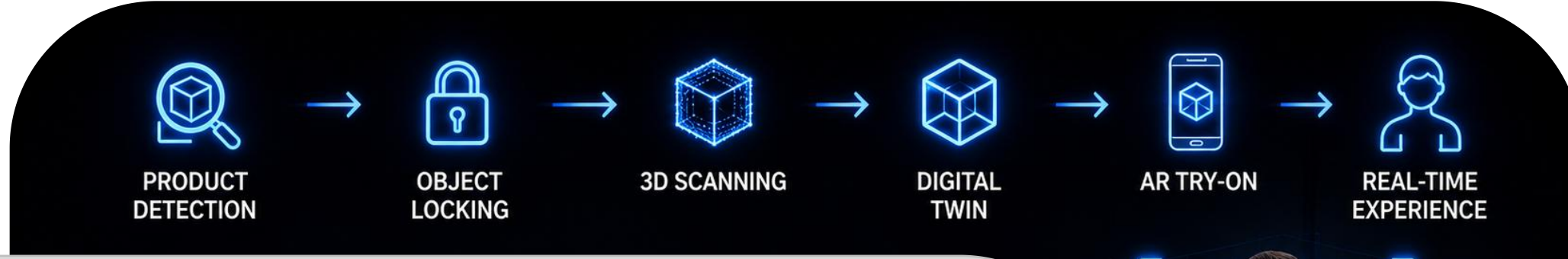


ZcannerReality: An AI and Augmented Reality-Based Universal Virtual Try-On Platform



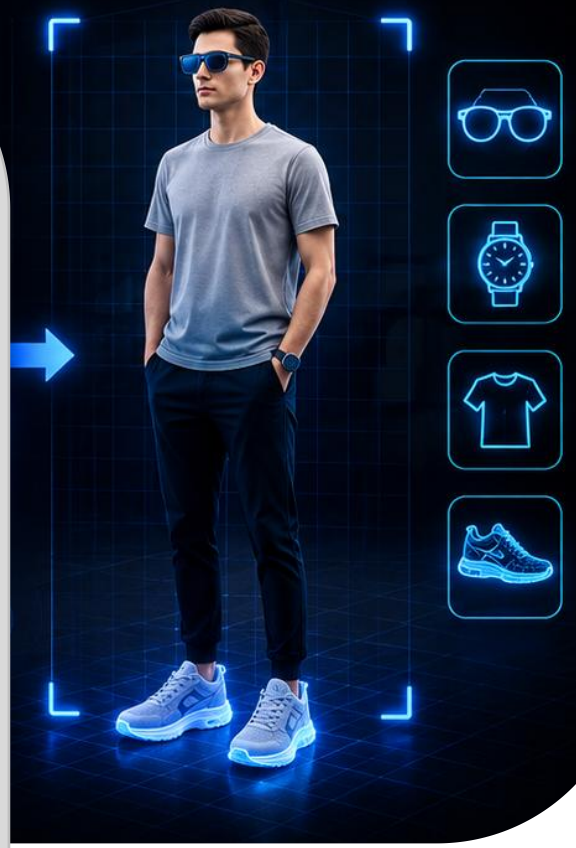
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ABSTRACT:

- **Universal AI Try-On**: ZcannerReality uses AI, Computer Vision, and AR to support clothing, shoes, watches, sunglasses, and other accessories across brands.
- **Smart Object Detection & Locking**: AI detects the product while users can mark its texture/area to lock, track, and isolate the exact object.
- **Fast Digital Twin Creation**: Predefined 3D models provide instant twins for known products, while AI reconstruction creates 3D models for unknown objects using depth, mesh, and texture generation.
- **Body Detection & Real-Time AR**: Computer Vision identifies the correct body part and AR accurately places and tracks the virtual product during real-time try-on.
- **Better Shopping Experience**: Users can visualize and interact with products before buying, improving purchase confidence and helping reduce product returns.





PRIVACY CONCERNS
During Physical Trials

LACK OF PRIVACY
Open areas or gaps in trial rooms can make you feel exposed.

UNWANTED ATTENTION
Crowded spaces or people nearby may cause discomfort.

LIMITED PERSONAL SPACE
Small or shared trial rooms do not provide enough personal space.

COMFORT MATTERS
Everyone deserves to try products in a comfortable and respectful environment.

Your comfort and privacy are important.
You deserve a better way to try.

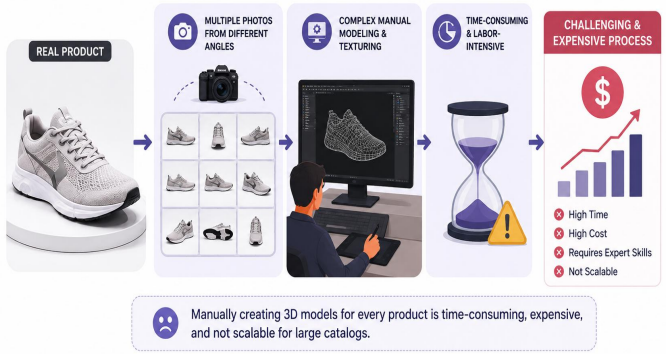
CUSTOMERS CANNOT VISUALIZE PRODUCTS BEFORE BUYING



LIMITED VIRTUAL TRY-ON SOLUTIONS
Existing solutions are restricted to specific brands or categories



DIFFICULTY CREATING 3D MODELS
OF REAL PRODUCTS



Manually creating 3D models for every product is time-consuming, expensive, and not scalable for large catalogs.

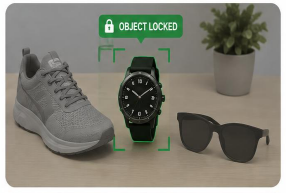
INCORRECT OBJECT SELECTION DURING SCANNING
Multiple objects in the camera view can confuse the system.

PROBLEM
The system may select the wrong object when multiple items are present.



Wrong selection leads to inaccurate 3D model creation and poor AR try-on experience.

SOLUTION
Users can mark and lock the exact object using texture/area-based object locking.



Correct object locked → Accurate 3D model → Better AR placement and try-on.

WHY IT MATTERS
Accurate object selection ensures the right product is scanned, modeled, and virtually tried on.

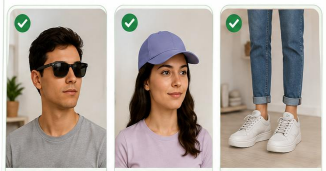
INACCURATE AR PLACEMENT
Virtual products may not be positioned on the correct body part, leading to an unrealistic try-on experience.

PROBLEM
AR places the product at the wrong location due to lack of body part detection.



Results in unrealistic try-on and poor user experience.

SOLUTION
AI detects and tracks body parts to place the product accurately in real time.



Accurate placement → Realistic try-on → Higher user satisfaction

WHY IT MATTERS
Correct AR placement ensures users see how the product actually looks on them, increasing confidence and reducing returns.

PROBLEM STATEMENT:

- **Limited Visualization**: Customers cannot preview products before buying.
- **High Returns**: Uncertainty about fit and appearance causes returns.
- **Limited Try-On**: Existing solutions support limited brands and categories.
- **3D Creation**: Manual 3D modeling is slow, costly, and difficult to scale.
- **Object Selection**: Multiple objects can cause incorrect scanning and locking.
- **AR Accuracy**: Poor body tracking can cause incorrect product placement.
- **Privacy Concerns**: Physical trial rooms may cause privacy and comfort issues.
- **No Unified Platform**: AI, 3D, and AR are rarely combined in one solution.

OBJECTIVES :

01 Develop Real-Time Product Detection

Detect and identify fashion products (clothing, shoes, watches, accessories, etc.) from real-world images or live camera feed.



02 Generate 3D Models from Detected Objects

Convert detected 2D product images into accurate 3D models using AI-based 3D reconstruction techniques.



03 Enable Object Placement in Augmented Reality

Allow users to place the generated 3D products in their real environment using AR with accurate scale, orientation, and positioning.



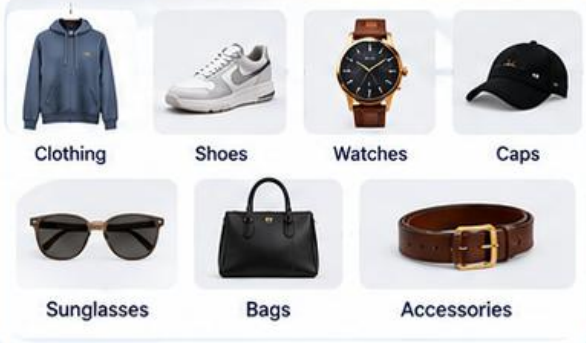
04 Provide Virtual Try-On for Fashion Products

Enable users to virtually try on clothing, shoes, and accessories using AI and AR for a realistic and interactive experience.



05 Support Multiple Product Categories

Build a system that works across different fashion categories such as clothing, shoes, watches, bags, sunglasses, and accessories.



06 Build a Unified and User-Friendly Platform

Integrate all functionalities (detection, 3D model generation, AR placement, and virtual try-on) into a single, easy-to-use application for an interactive and seamless user experience.



LITERATURE SURVEY:

S.No.	Author / Title	Methodology	Limitations	Our Contribution to Overcome the Limitations
1.	Zhao et al. – M3D-VTON: Monocular-to-3D Virtual Try-On (2021)	Uses monocular depth prediction and texture fusion to generate a 3D try-on model from limited visual input.	Mainly focused on clothing and does not provide a universal multi-category AR platform.	Extend the concept to shoes, watches, sunglasses, accessories and general objects with a common digital-twin pipeline.
2.	Cheng et al. – YOLO-World: Real-Time Open-Vocabulary Object Detection (2024)	Uses vision-language modeling and text prompts for open-vocabulary object detection; reported 35.4 AP on LVIS at ~52 FPS.	Provides detection only; it does not generate digital twins or perform AR try-on.	Connect detection to object locking, 3D twin retrieval/ reconstruction, and AR placement.
3.	Kirillov et al. – Segment Anything (2023)	Promptable, zero-shot segmentation for isolating arbitrary objects; trained using the SA-1B dataset.	Segmentation alone does not create a 3D model or digital twin and may require user refinement.	Add user-guided texture/area marking and object locking, then send the isolated object into tyie 3D pipeline.
4.	Liu et al. – Grounding DINO: Open-Set Object Detection (2023)	Uses vision-language detection to locate objects from natural-language prompts; reported 52.5 AP on zero-shot COCO.	Detection must be combined with segmentation and reconstruction for a complete product digitization workflow.	Use open-vocabulary detection as the front-end selection stage of a complete detection-to-digital-twin pipeline.
5.	Zhu et al. – DeepFashion3D / Fashion3D Dataset (2020)	Provides 3D scans of garments such as shirts, pants and dresses for 3D fashion research.	Existing fashion datasets are fragmented and do not combine broad product categories with paired 3D and try-on data.	Build a multi-category workflow and in-house dataset/captures covering fashion and general objects.

RESEARCH GAP:

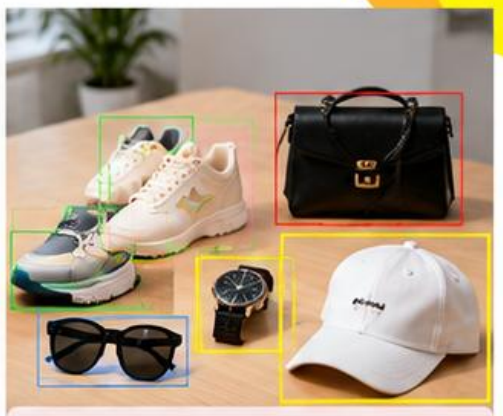
- **Limited Scope:** Existing research mainly focuses on clothing try-on and supports only a few brands and categories.
- **Dependence on Manual / Specialized Scanning:** Many methods require multi-view scanning or specialized hardware, making 3D model creation time-consuming and costly for consumer use.
- **Lack of Robust Object Selection:** Existing approaches do not provide a reliable method to detect, lock, and isolate the exact object when multiple items are present in a scene.
- **Limited Integration:** AI detection, 3D model generation, and AR try-on are usually developed separately, with no unified end-to-end platform.
- **Accuracy and Real-Time Performance:** Current solutions often face challenges in accurate body tracking and realistic product placement during real-time AR try-on.
- **Limited to Known Products:** Many systems work only with predefined product models and do not support unknown objects or new categories efficiently.



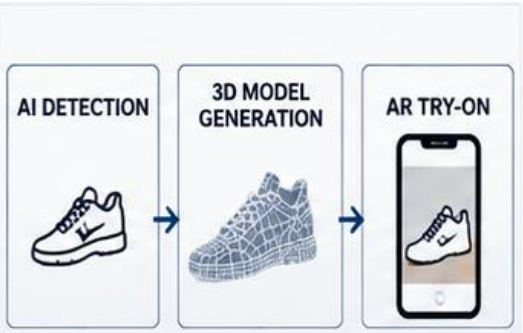
Only clothing focused



Requires specialized scanning setup



Multiple objects cause incorrect selection



Developed separately, no unified platform



Inaccurate tracking and placement



Not suitable for unknown products

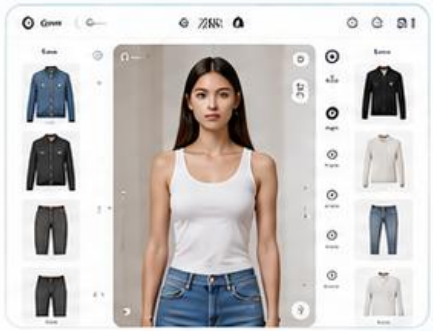
EXISTING SYSTEMS :

01 Web / App Based Virtual Try-On

ZARA

H&M

- Allows users to virtually try on clothing using 2D images or AI-based models.
- Supports limited brands and categories (mainly top wear and bottom wear).
- Works on mobile apps or websites, no real 3D modeling.



02 Dedicated 3D Scanning Systems (for Product Digitization)

shining 3D

Artec 3D

- Uses specialized 3D scanners to create high-quality 3D models of physical objects.
- Produces accurate size, shape and texture models.
- Requires expensive hardware and controlled environment.



03 Object Recognition to AR (Mobile Apps)

Google Lens

IKEA Place

- Detects real-world objects using the phone camera.
- Allows users to place similar 3D models of the object in AR.
- Mainly used for furniture and home decor items.
- Relies on pre-built 3D models available in the app database.



EXISTING SYSTEMS (TRY-ON & OBJECT TO AR)

04 E-commerce AR Visualization

amazon AR View

Flipkart View in Your Room

- Allows users to place product (furniture, electronics, etc.) in their real environment using AR.
- Provides realistic size and placement preview before purchase.
- Works only for selected products with available 3D models.



05 AR Measuring / Scanning Apps

Apple Measure

Google Measure

- Uses camera and depth sensors to measure real-world objects.
- Creates basic 3D models or dimensions of the object.
- Useful for measuring furniture, room size, or small objects.
- Provides simple AR visualization (no detailed 3D models).

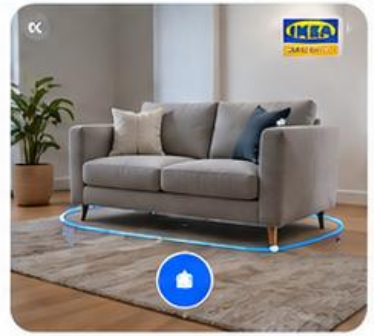


06 Furniture and Interior AR Apps

IKEA Place

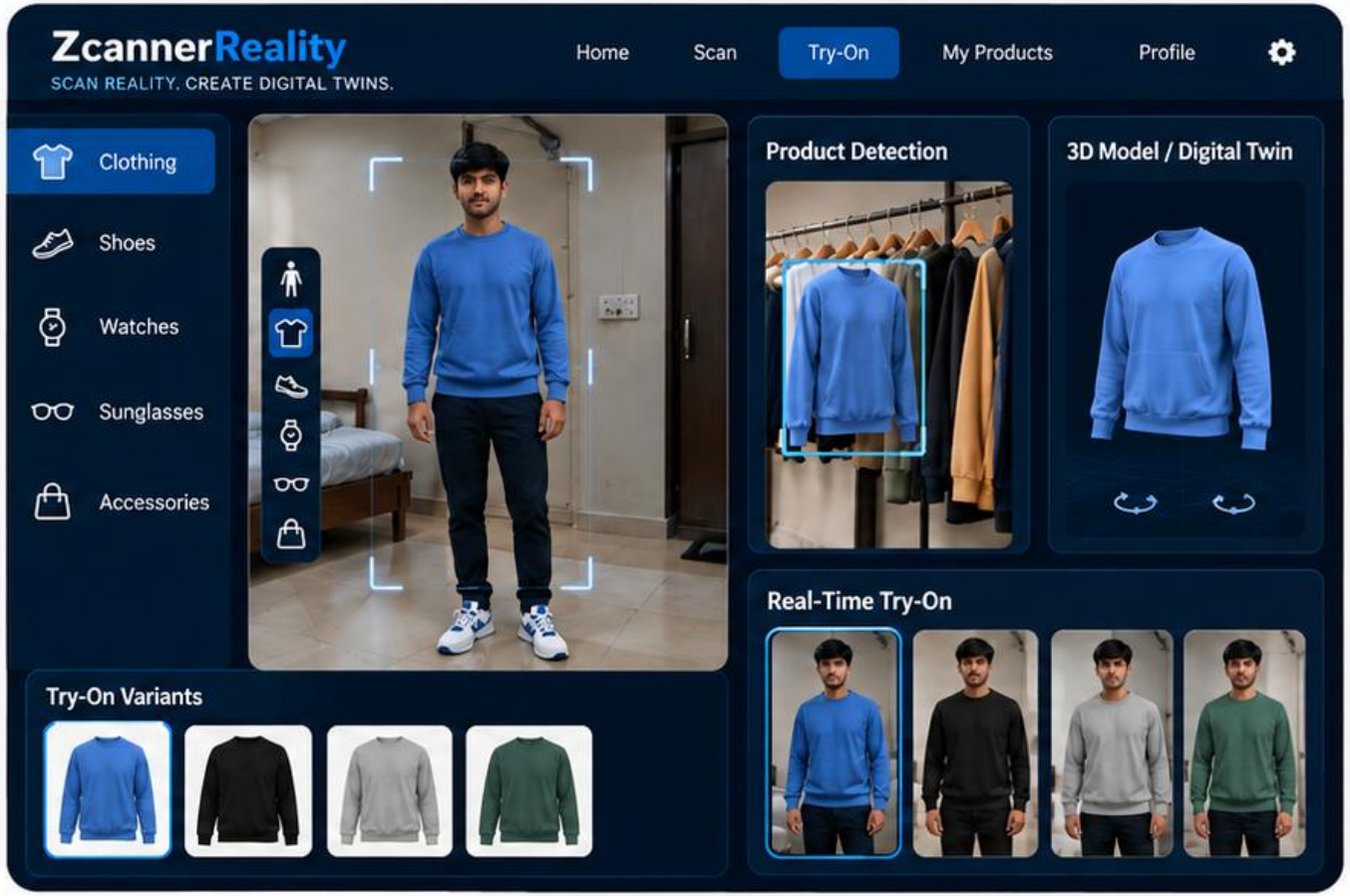
Wayfair View in Room

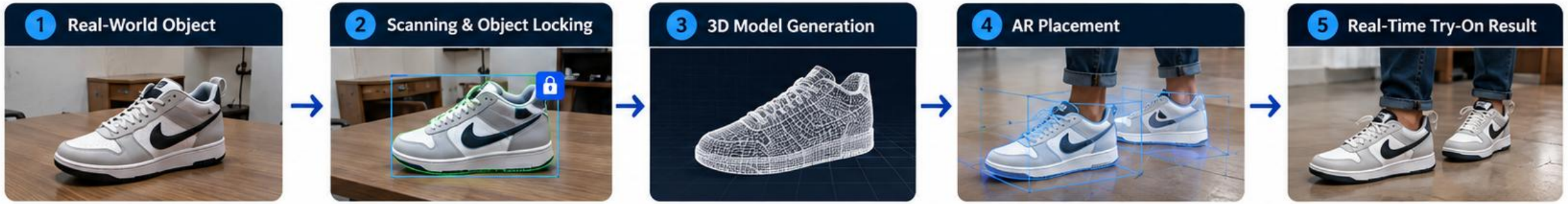
- Allows users to place furniture and decor items in AR.
- Uses pre-designed 3D models from the company catalog.
- Helps users visualize size, design and fit in their own space.
- Limited to available catalog items and specific product categories.



OUR SOLUTION:

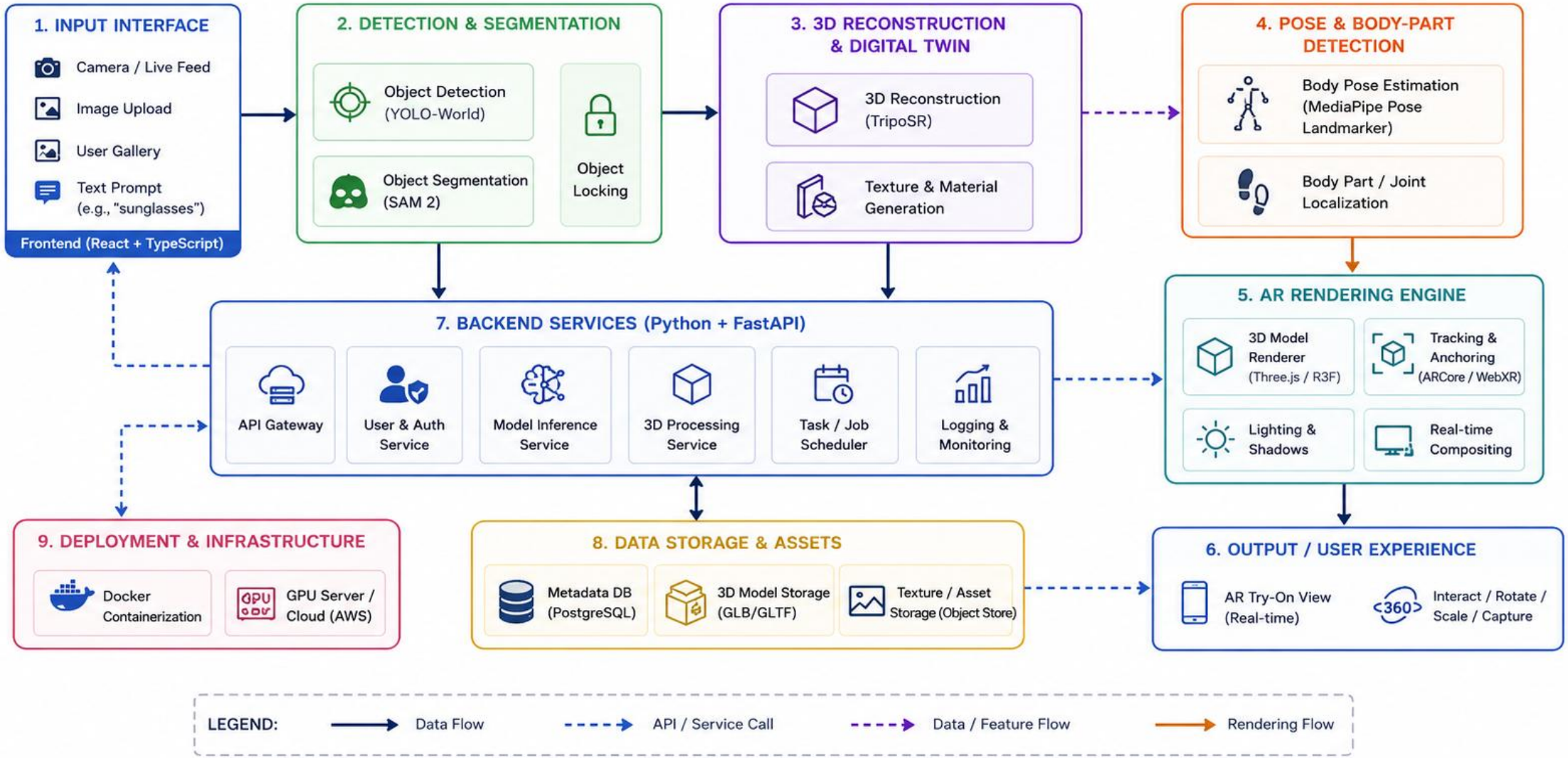
- **AI Detection:** Detects and identifies fashion products automatically.
- **Object Locking:** Marks and locks the exact object for accurate scanning.
- **Fast Digital Twin:** Uses predefined 3D models for instant results.
- **AI 3D Reconstruction:** Creates 3D models for unknown objects.
- **Body Detection:** Detects body parts for accurate product placement.
- **Real-Time AR:** Places and tracks products realistically in AR.
- **Privacy-Friendly:** Enables virtual trials without physical changing rooms.
- **Universal Platform:** Combines AI + 3D + AR for multiple categories.





ARCHITECTURE DIAGRAM

ZCANNERREALITY SYSTEM ARCHITECTURE (MODULE DIAGRAM)



METHODOLOGY:

01 Data Collection

- Collect real-world images of fashion products (clothing, shoes, accessories, furniture).
- Use open datasets and custom images.
- Capture images from different angles and backgrounds.



02 AI-based Product Detection

- Use YOLOv8 (or YOLO-World) to detect and classify fashion products from images or live camera feed.
- Identify product category (clothing, shoes, accessories, furniture).
- Generate bounding boxes and labels.



03 3D Model Reconstruction

- Convert detected 2D images into 3D models using AI-based 3D reconstruction (e.g., InstantMesh / TripoSR / similar models).
- Generate textured and realistic 3D models of the products.



2D Image (Detected Object) → 3D Model (Generated) → Textured 3D Model (Realistic)

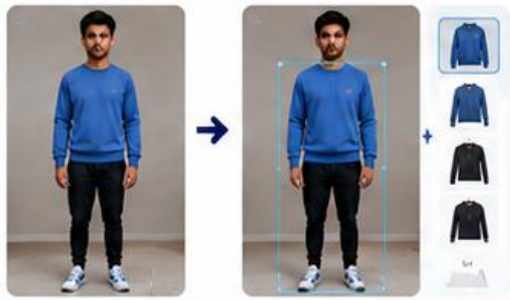
04 Object Placement in AR

- Use AR framework (e.g., ARCore/ ARKit/Unity) to place the 3D model in the real-world environment.
- Allow users to move, rotate and scale the object.



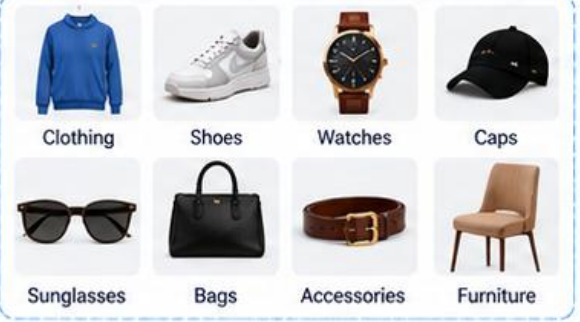
05 Virtual Try-On (for Clothing)

- Use a human pose estimation model to detect body keypoints.
- Map the 3D clothing model onto the user's body in real-time.
- Provide realistic visualization



06 Multi-Category Support

- The same pipeline works for clothing, shoes, watches, bags, sunglasses, furniture, etc.



07 Real-Time Interaction

- Allow users to interact with the placed/try-on products in real time.
- Support change of color, size, style and different brands.



08 Output and User Experience

- Display realistic 3D models and AR placement in a unified platform.
- Provide easy-to-use interface for scanning, try-on and object placement.
- Enable users to visualize and interact with fashion products in real time.



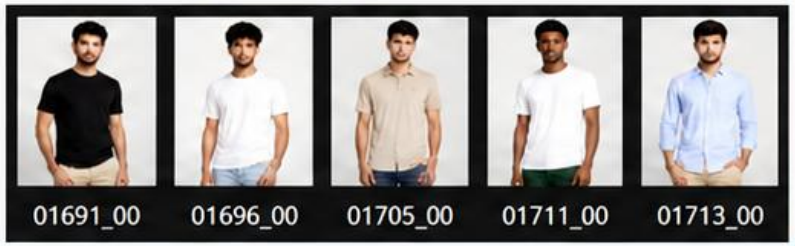
TECH STACK:

TECH STACK	PURPOSE / USAGE
React + Vite + TypeScript	• Main ZcannerReality app UI
Three.js + React Three Fiber + Drei	• Display, rotate, scale and interact with 3D digital twins. React Three Fiber is specifically a React renderer for Three.js.
YOLO-World	• Detect products such as shoes, watches, glasses, clothes, etc. using open-vocabulary text prompts.
SAM 2	• Precisely isolate the selected object and track it through camera frames; SAM 2 supports image/video segmentation and interactive prompts.
TripoSR	• Generate a 3D model from a single product image for unknown objects. Fast single-image 3D reconstruction.
MediaPipe Pose Landmarker	• Detect body landmarks in images/video and provide 3D world coordinates for positioning products.
ARCore / WebXR	• Track the real environment and anchor 3D objects in AR. Provides motion tracking, environmental understanding and light estimation with WebXR support for web AR.
Python + FastAPI	• Connect React with the AI pipeline and handle image/model processing requests. Auto docs and OpenAPI support.
PyTorch	• Run and integrate the detection, segmentation and reconstruction models.
GLB / glTF	• Store and transfer generated/predefined digital twins efficiently.
PostgreSQL + Object Storage	• Store product information, model metadata and .glb digital twins.
OpenCV	• Camera frames, image preprocessing, cropping and computer-vision utilities.
Docker + GPU Server	• Package and run the AI backend consistently with GPU acceleration.

DATASET :

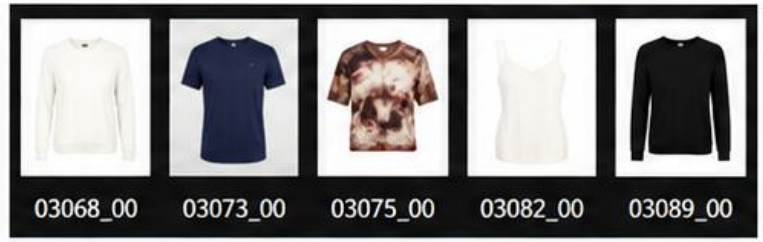
01 Person Images (Male)

Male human images used as the base person input.



02 Clothing Images

Target garment images (tops, t-shirts, shirts, etc.) used for virtual try-on.



03 Clothing Masks

Binary masks to isolate the garment from the background.



04 Clothing-Agnostic Images

Person images with upper clothing removed (body structure).



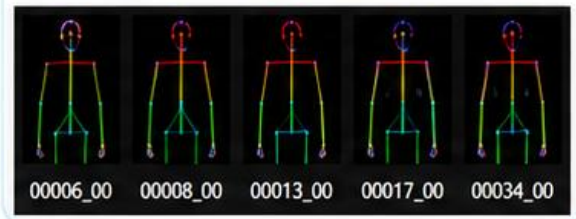
05 Human Parsing

Semantic labels for body parts (head, arms, torso, legs, etc.).



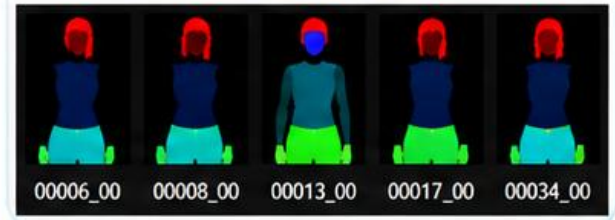
06 Pose / Keypoints

Body keypoints / skeleton for garment alignment.



07 Pose Rendering / DensePose

Dense body representation for realistic warping and placement.



08 Complete Dataset Overview (VITON-HD)

A high-resolution virtual try-on dataset that includes all the above components (person images, clothing images, masks, clothing-agnostic images, parsing, keypoints, and DensePose) for training and evaluation.



IMPLEMENTATION :

01 Input Image & Processing

- Take user image (webcam/upload).
- Preprocess the image (resize, normalization).



Input Person Image

02 Human Parsing

- Segment body parts (head, arms, torso, legs).
- Helps to align garments accurately.



Parsing Result

03 Garment Detection

- Detect and classify clothing items using YOLOv8 (e.g., t-shirt, shirt, jacket).



Detected Garment Images

04 3D Model Generation

- Convert 2D garment images to 3D models using AI-based 3D reconstruction (e.g., InstantMesh / TripoSR).



2D Garment Image

Generated 3D Model

05 Virtual Try-On (Clothing)

- Use a try-on model to warp and fit the 3D garment to the user's image.
- Generate a realistic try-on result.



Input Person

Try-On Result

06 Object Detection for AR

- Detect real-world objects (shoes, bags, watches, furniture, etc.) using YOLOv8-World.
- Identify object category and location.



07 AR Placement and Visualization

- Place and render the selected 3D object in real world using AR (e.g., ARCore / ARKit / Unity).
- Allow users to move, rotate and scale the object.

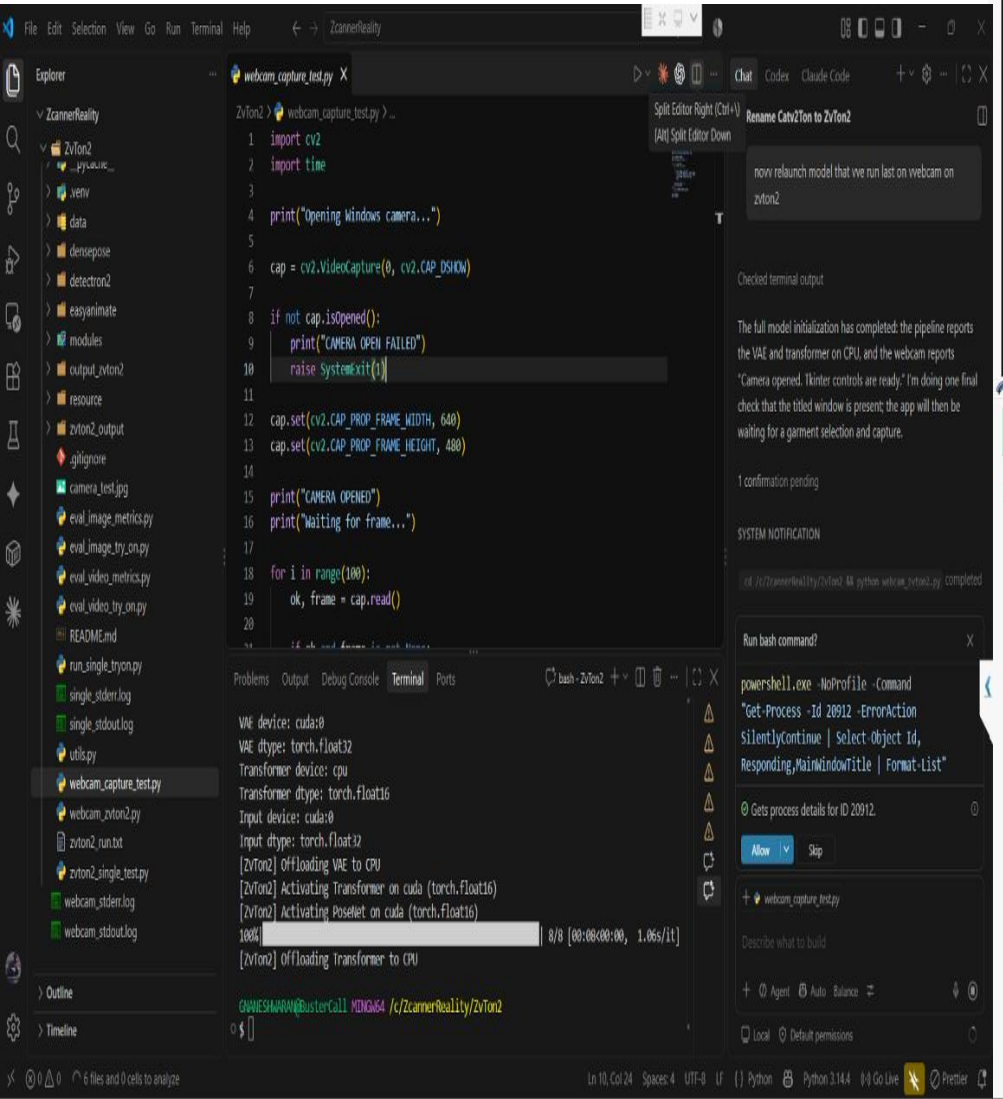


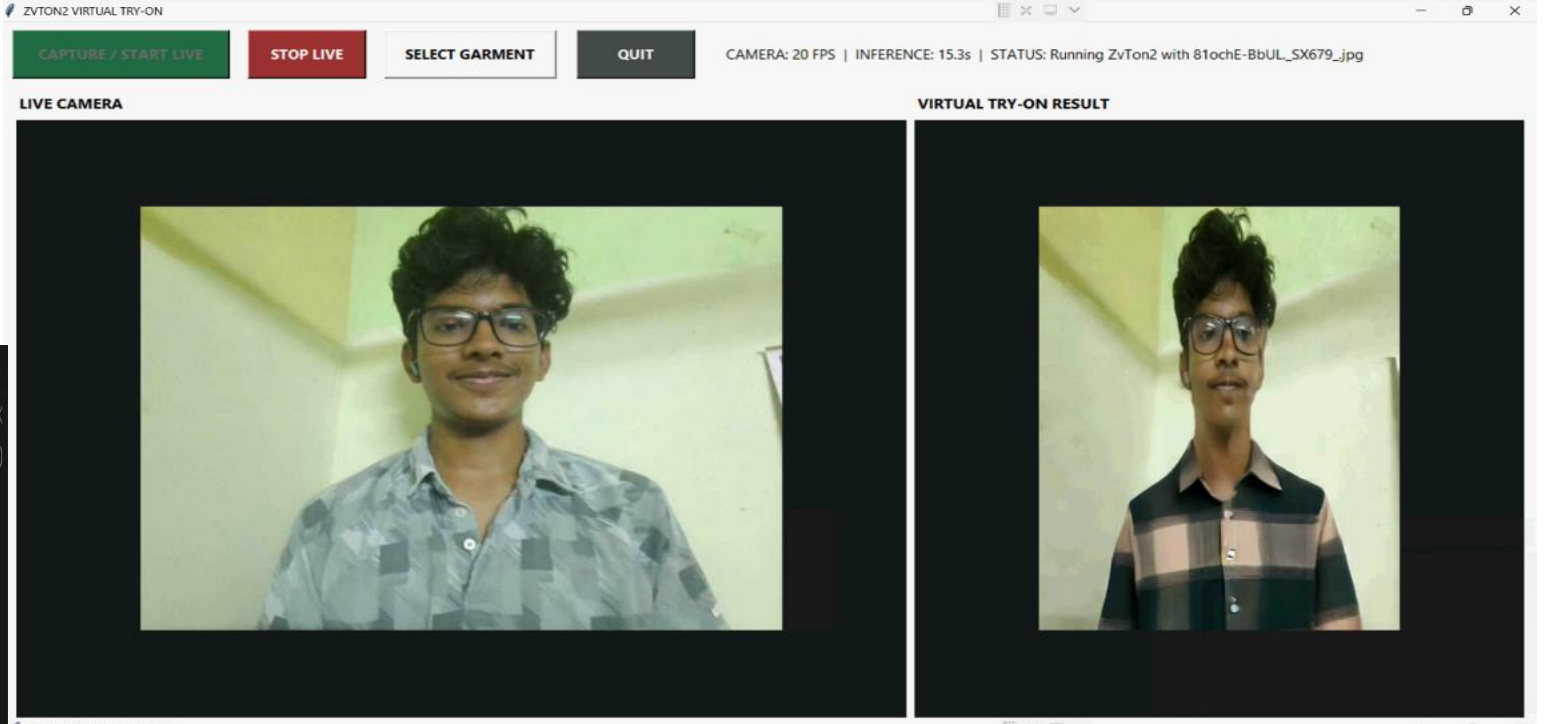
08 Output and User Experience

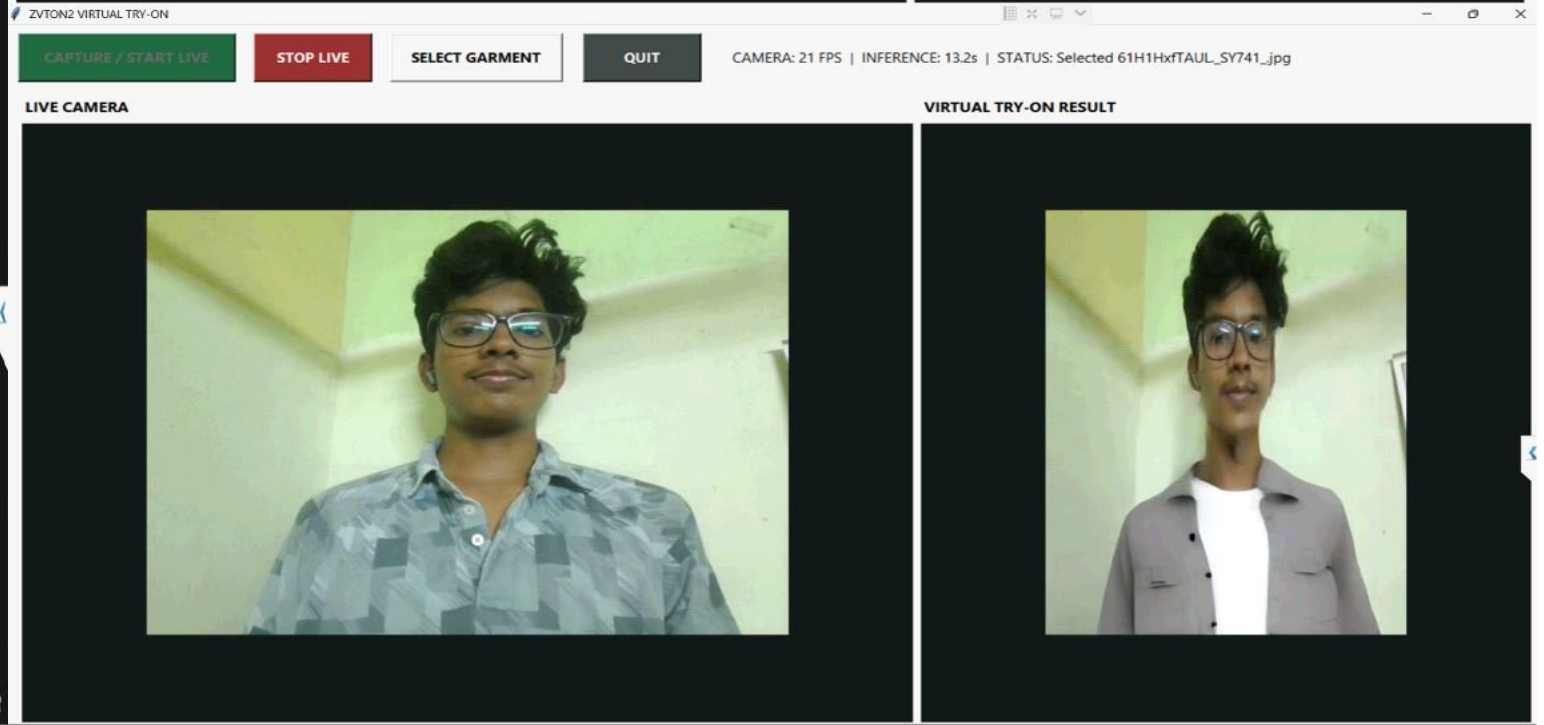
- Display virtual try-on and AR placed objects in a unified system.
- Provide an easy-to-use interface for scanning, try-on and object placement.



RESULT:


webcam_capture_test.py
import cv2
import time
print("Opening Windows camera...")
cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)
if not cap.isOpened():
print("CAMERA OPEN FAILED")
raise SystemExit(1)
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
print("CAMERA OPENED")
print("Waiting for frame...")
for i in range(100):
ok, frame = cap.read()
VAE device: cuda:0
VAE dtype: torch.float32
Transformer device: cpu
Transformer dtype: torch.float16
Input device: cuda:0
Input dtype: torch.float32
[ZvTon2] Offloading VAE to CPU
[ZvTon2] Activating Transformer on cuda (torch.float16)
[ZvTon2] Activating PoseNet on cuda (torch.float16)
100% 8/8 [00:08<00:00, 1.06s/it]
[ZvTon2] Offloading Transformer to CPU


ZVTON2 VIRTUAL TRY-ON
CAPTURE / START LIVE
STOP LIVE
SELECT GARMENT
QUIT
CAMERA: 20 FPS | INFERENCE: 15.3s | STATUS: Running ZvTon2 with 81ochE-BbUL_SX679_jpg
LIVE CAMERA
VIRTUAL TRY-ON RESULT


ZVTON2 VIRTUAL TRY-ON
CAPTURE / START LIVE
STOP LIVE
SELECT GARMENT
QUIT
CAMERA: 21 FPS | INFERENCE: 13.2s | STATUS: Selected 61H1HxfTAUL_SY741_jpg
LIVE CAMERA
VIRTUAL TRY-ON RESULT

FUTURE ENHANCEMENT :

01 More Clothing Categories

- Extend support to a wider range of clothing items such as pants, dresses, jackets, traditional wear and accessories.



02 Improved 3D Generation

- Use advanced and lightweight 3D reconstruction models to generate more accurate, high-quality and realistic 3D models of garments and users.



2D Garment Image → High-Quality 3D Model

03 Better FIT Estimation

- Incorporate body measurements and size prediction to provide personalized and realistic fitting results for different body shapes.



Height: 178 cm
Chest: 96 cm
Waist: 82 cm
Size: M (Recommended)

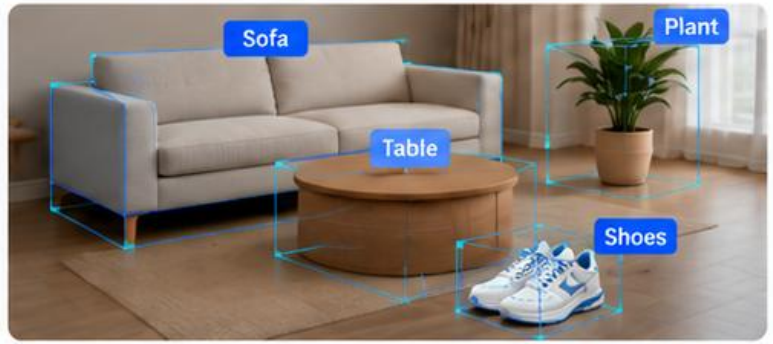
04 Real-time AR Experience

- Develop a more interactive and realistic AR environment with real-time rendering, lighting adjustment and multi-object placement.



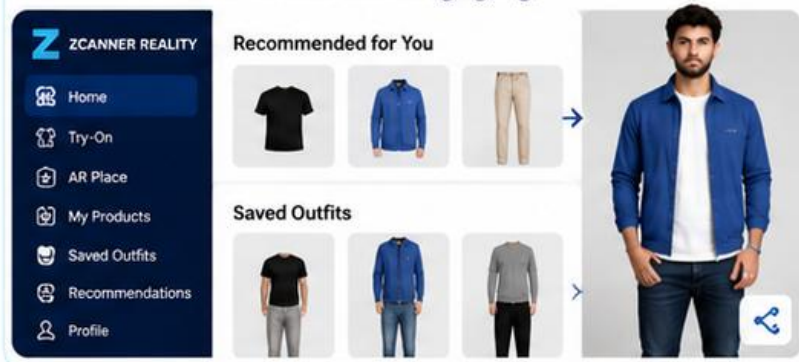
05 Multi-Object AR Placement

- Allow users to place multiple objects (furniture, accessories, footwear, etc.) in the real world for complete virtual shopping experience.



06 Enhanced User Experience

- Add features like user accounts, saved outfits, AI-based style recommendations and social sharing to make the system more useful and engaging.



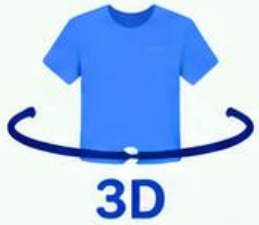
CONCLUSION :

01 What We Did



We developed **Zcanner Reality**, a system that combines deep learning and 3D modeling to create a **virtual try-on and object-to-AR ecommerce platform.**

02 What It Does



It converts 2D images of clothing into realistic 3D models, allows users to **virtually try on clothes**, and lets them place products in their **real environment using AR.**

03 For Whom



It is useful for **online shoppers, fashion enthusiasts**, and **e-commerce businesses** who want a more interactive, realistic and confident shopping experience.

04 Impact



It helps **reduce uncertainty** in online shopping, **saves time and cost, increases customer satisfaction**, and **supports sustainable and efficient retail** through digital visualization.





In conclusion, **Zcanner Reality** bridges the gap between online and real-world shopping by bringing **realistic visualization, interactivity and convenience** to everyone, making e-commerce more **engaging, confident and accessible.**

GANTT CHART – ZCANNERREALITY PROJECT TIMELINE

Project Duration: **July 1, 2026 – October 31, 2026**

Task / Activity	Duration (Days)	JULY 2026	AUGUST 2026	SEPTEMBER 2026	OCTOBER 2026
1. Requirement Analysis & Project Planning	7	Jul 1 – Jul 7			
2. Literature Survey & Existing System Study	10	Jul 8 – Jul 17			
3. System Architecture & Module Design	12	Jul 18 – Jul 29			
4. Dataset Collection & Preparation	14	Jul 25 – Aug 7			
5. Object Detection Model Development	18		Aug 1 – Aug 18		
6. Human Parsing & Pose Module	16		Aug 10 – Aug 25		
7. Virtual Try-On / Garment Processing	20		Aug 20 – Sep 8		
8. 3D Reconstruction & Digital Twin Module	22			Sep 1 – Sep 22	
9. AR Object Detection & Placement Integration	20			Sep 15 – Oct 4	
10. System Integration, Testing & Optimization	20			Sep 25 – Oct 14	
11. Final Evaluation, Documentation & Presentation	17				Oct 15 – Oct 31

LEGEND: Planning · Research · Design · Data · Detection · Pose / Parsing · Virtual Try-On · 3D / Digital Twin · AR Integration · Testing · Finalization

REFERENCES:

- M3D-VTON – Monocular-to-3D SVirtual Try-On Network (ICCV 2021)
  Supports your 3D Fashion Twin / virtual try-on concept.
  https://openaccess.thecvf.com/content/ICCV2021/html/Zhao_M3D-VTON_A_Monocular-to-3D_Virtual_Try-On_Network_ICCV_2021_paper.html?
- YOLO-World – Real-Time Open-Vocabulary Object Detection (CVPR 2024)
  Supports your AI product detection and multi-category detection approach.
  https://openaccess.thecvf.com/content/CVPR2024/html/Cheng_YOLO-World_Real-Time_Open-Vocabulary_Object_Detection_CVPR_2024_paper.html?
- Segment Anything (SAM) (ICCV 2023)
  Supports your object selection, segmentation and object-locking concept.
https://openaccess.thecvf.com/content/ICCV2023/html/Kirillov_Segment_Anything_ICCV_2023_paper.html?

